

BIVALVES

Highly specialized aquatic molluscs, bivalves are instantly recognizable, with a hinged shell that opens up to access food and oxygen-rich water.

Bivalves are identified from their shell, which consist of two plates, called valves, joined together by a hinge. To protect against predators and the dangers of drying out, the shell clamps tightly shut by contracting powerful muscles, sealing most or all of the body inside. Seashore species are regularly exposed by receding tides.

Within the class there is a huge variation in size, ranging from the tiny fingernail clam at about 6 mm (¼ in) to the giant clam, which can grow to 1.4 m (4½ ft) across. Some bivalves attach to rocks and hard surfaces by a bundle of tough threads,

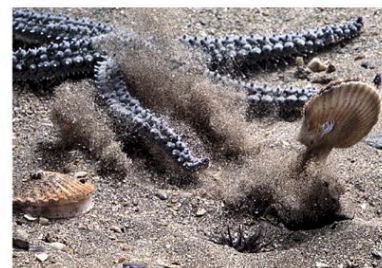
called a byssus. Others use a well-developed muscular foot for burrowing into muddy sediment. A few bivalves, such as scallops, swim freely, using a method of jet propulsion.

Water is pumped in and out of the shell through tubes, called siphons. The water supplies oxygen and food, which are collected as the water passes over modified gills. Food particles caught in the sticky mucus around the gills are wafted towards the mouth by microscopic hairs.

BIVALVES AND PEOPLE

Mussels, clams, and oysters are important sources of food, and some oysters create fine quality pearls by coating a foreign object with layers of nacre. Bivalves are also useful indicators of water quality as many cannot survive high levels of pollution.

PHYLUM	MOLLUSCA
CLASS	BIVALVIA
ORDERS	19
FAMILIES	105
SPECIES	9,733



A queen scallop claps the two valves of its shell together to accelerate away from the unwelcome attention of a predatory starfish.

OYSTERS AND SCALLOPS

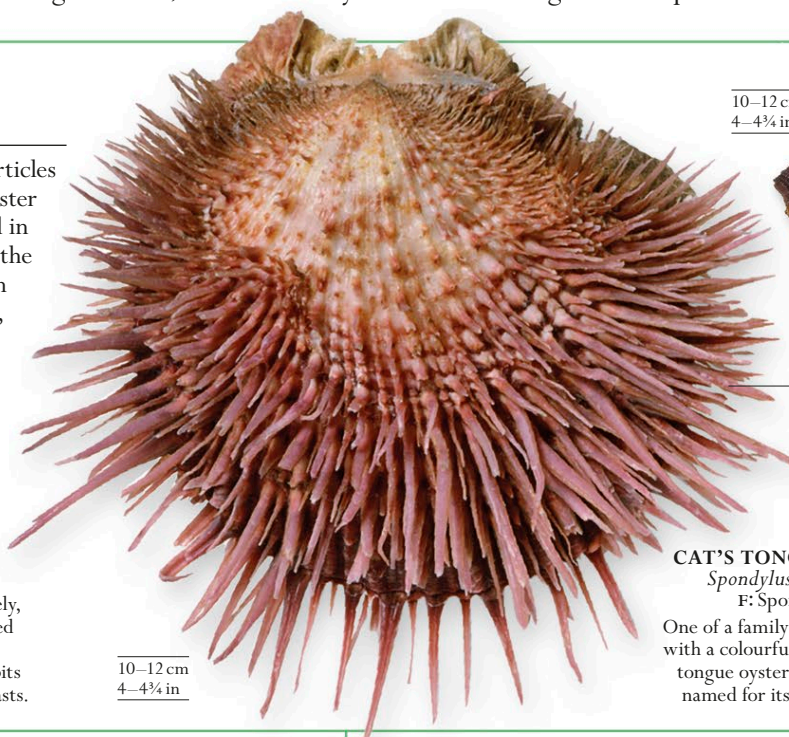
The Ostreoida feed on tiny food particles filtered from the seawater. Many oyster species live permanently submerged in coastal waters, fixed to the rock by the gland that produces byssus thread in other bivalves. Scallops, by contrast, can swim freely by clapping their valves open and shut.



12–15 cm
4¾–6 in

GREAT SCALLOP
Pecten maximus
F: Pectinidae

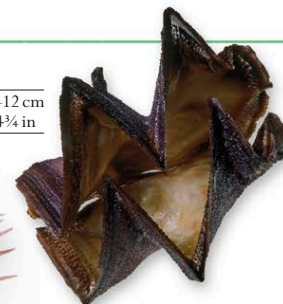
As well as swimming freely, scallops have jet-propelled escape responses. This commercial species inhabits fine sands of European coasts.



10–12 cm
4–4¾ in

CAT'S TONGUE OYSTER
Spondylus linguafelis
F: Spondylidae

One of a family of thorny oysters with a colourful mantle, the cat's tongue oyster of the Pacific is named for its spiny exterior.



COCK'S COMB OYSTER
Lopha cristagalli
F: Ostreidae

Close relatives of scallops, many species of oyster are prized as food and for their pearls. This is an Indo-Pacific species.



8–10 cm
3¼–4 in

EDIBLE OYSTER
Ostrea edulis
F: Ostreidae

Formerly abundant in Europe, this commercially important species is now over-fished in some areas. The species is inedible during summer months when breeding.

ARC CLAMS AND RELATIVES

The shell of these bivalves is closed by two strong muscles, and hinge along a straight line bearing a continuous series of teeth. Like their relatives, the Arcoida have a reduced foot and large gills that help trap food particles.



5–7 cm
2–2¾ in

NOAH'S ARK
Arca noae
F: Arcidae

Arc shells are square-edged, thick-shelled bivalves. Noah's Ark is a byssus-attached intertidal species on rocky coasts of the E. Atlantic.



5–6 cm
2–2¼ in

EUROPEAN BITTERSWEET
Glycymeris glycymeris
F: Glycymerididae

Dog shells are rounded relatives of arc shells. This N.E. Atlantic species, fished in Europe, has a sweet flavour but is tough when overcooked.

MARINE MUSSELS

The Mytiloida have a distinctively shaped elongate, asymmetrical shell that attaches to rocks by byssus threads. Only one of their two shell-closing muscles is well developed.



8–10 cm
3¼–4 in

COMMON MUSSEL
Mytilus edulis
F: Mytilidae

The most commercially important mussel in Europe, this long-lived species occurs in dense beds. It can tolerate the low salinity of estuarine waters.

FAN MUSSELS AND RELATIVES

Members of the Pterioda include long-hinged, winged oysters and T-shaped hammer-oysters as well as fan mussels. The group also contains commercially important marine pearl oysters.



25–40 cm
10–16 in

FLAG FAN MUSSEL
Atrina vexillum
F: Pinnidae

Fan mussels, such as this species from western European coastlines, have a triangular shell attached by a byssus in soft sediment.